Chapter 3
Motion in Central Potential $V(r)$

3.1 Angular momentum and the Lagrangian

$$\vec{L} = \mu r^2 \dot{\phi}\,\hat{z}, \qquad \vec{r} = r\,\hat{r}, \qquad \vec{p} = \mu\dot{\vec{r}}, \tag{3.1}$$

with $\vec{L}$ a constant of the motion. The Lagrangian is

$$L = T - V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - V(r). \tag{3.2}$$

Since ϕ is cyclic,

$$\frac{\partial L}{\partial \phi} = 0 \quad\Longrightarrow\quad p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \mu r^2 \dot{\phi} \equiv \ell \quad \text{conserved.} \tag{3.3}$$

The effective potential and the energy are

$$V_{\text{eff}}(r) = V(r) + \frac{\ell^2}{2\mu r^2}, \tag{3.4}$$

$$E = T + V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = \frac{\mu}{2}\dot{r}^2 + V_{\text{eff}}(r). \tag{3.5}$$

3.2 Universal method $\Rightarrow$ complete solution

$$\boxed{\begin{aligned} t &= \int_{r_0}^{r} \frac{\mathrm{d}r'}{\sqrt{\dfrac{2}{\mu}\left(E - V(r') - \dfrac{\ell^2}{2\mu r'^2}\right)}} \\[2ex] \phi &= \phi_0 + \int_{t_0}^{t} \frac{\ell}{\mu r^2(t')}\,\mathrm{d}t' \end{aligned}} \qquad \begin{aligned} t &= t(r) \\[6ex] \phi &= \phi(t) \end{aligned} \tag{3.6}$$

3.3 Differential equation for $r = r(\phi)$

$$\ell = \mu r^2 \dot{\phi} \quad\Longrightarrow\quad \dot{\phi} = \frac{\ell}{\mu r^2} = \frac{\mathrm{d}\phi}{\mathrm{d}t}, \tag{3.7}$$

so that for any F,

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\mathrm{d}F}{\mathrm{d}\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t} = \underbrace{\frac{\ell}{\mu r^2}\frac{\mathrm{d}F}{\mathrm{d}\phi}}. \tag{3.8}$$

The orbit is sketched in Fig. 3.1.

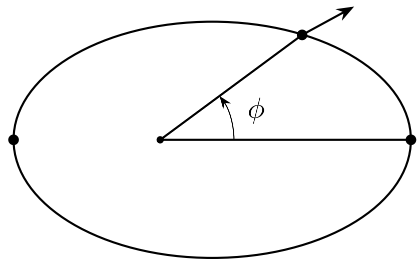

Figure 3.1: The orbit in the plane of motion.

With $-\frac{\partial V}{\partial r} = F(r)$, the radial equation of motion is

$$\mu\ddot{r} = -\frac{\partial V_{\text{eff}}}{\partial r} = -\frac{\partial}{\partial r}\left(V(r) + \frac{\ell^2}{2\mu r^2}\right), \tag{3.9}$$

and since $\frac{\partial}{\partial r}\left(r^{-2}\right) = (-2)r^{-3}$,

$$\mu\ddot{r} = F(r) + \frac{\ell^2}{\mu r^3}. \tag{3.10}$$

3.4 The substitution $u = 1/r$

Set

$$u = \frac{1}{r}, \qquad \frac{\mathrm{d}u}{\mathrm{d}\phi} = \frac{\mathrm{d}u}{\mathrm{d}r}\,\frac{\mathrm{d}r}{\mathrm{d}\phi} = -\frac{1}{r^2}\,\frac{\mathrm{d}r}{\mathrm{d}\phi}. \tag{3.11}$$

Writing F in u as $F(1/u)$ (Sec. 3.5),

$$\frac{\ell^2}{\mu r^3} = \frac{\ell^2}{\mu}u^3 \quad \Longrightarrow \quad \text{RHS: } F(1/u) + \frac{\ell^2}{\mu}u^3. \tag{3.12}$$

LHS: applying Eq. (3.8) twice,

$$\mu\,\frac{\mathrm{d}}{\mathrm{d}t}\,\frac{\mathrm{d}r}{\mathrm{d}t} = \mu\,\frac{\mathrm{d}}{\mathrm{d}t}\,\frac{\ell}{\mu r^2}\,\frac{\mathrm{d}r}{\mathrm{d}\phi} = \cancel{\mu}\,\frac{\ell}{\cancel{\mu} r^2}\,\frac{\mathrm{d}}{\mathrm{d}\phi}\left(\frac{\ell}{\mu r^2}\,\frac{\mathrm{d}r}{\mathrm{d}\phi}\right). \tag{3.13}$$

LHS:

$$\mu\ddot{r} = -\frac{\ell^2}{\mu}\,\frac{1}{r^2}\,\frac{\mathrm{d}}{\mathrm{d}\phi}\,\frac{\mathrm{d}u}{\mathrm{d}\phi} = -\frac{\ell^2}{\mu}\,u^2\,\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2}. \tag{3.14}$$

LHS=RHS

$$\boxed{-\frac{\ell^2}{\mu}\,u^2\,\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2} = F(1/u) + \frac{\ell^2}{\mu}u^3} \tag{3.15}$$

generic equation of trajectory for arbitrary $V(r)$

3.5 Force and potential in terms of u

$$F(r) = -\frac{\partial V}{\partial r}, \tag{3.16}$$

$$\frac{\mathrm{d}V}{\mathrm{d}r} = \frac{\mathrm{d}V}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}r} = -u^2\frac{\mathrm{d}V}{\mathrm{d}u}, \qquad \frac{\mathrm{d}u}{\mathrm{d}r} = -\frac{1}{r^2} = -u^2, \tag{3.17}$$

so that

$$F(1/u) = F(r) = -\frac{\mathrm{d}V}{\mathrm{d}r} = u^2\frac{\mathrm{d}V(1/u)}{\mathrm{d}u}, \tag{3.18}$$

$$-F(1/u) = -u^2\frac{\mathrm{d}V(1/u)}{\mathrm{d}u}. \tag{3.19}$$

Rewriting Eq. (3.15),

$$-F(1/u) = \frac{\ell^2}{\mu}u^2\left[u + \frac{\mathrm{d}u}{\mathrm{d}\phi^2}\right], \tag{3.20}$$

and equating the two expressions for $-F(1/u)$,

$$-\cancel{u^2}\frac{\mathrm{d}V}{\mathrm{d}u} = \frac{\ell^2}{\mu}\cancel{u^2}\left[u + \frac{\mathrm{d}^2u}{\mathrm{d}\phi^2}\right]. \tag{3.21}$$

3.6 Equation for the trajectory $r(\phi)$

$$\boxed{\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2} + u = -\frac{\mu}{\ell^2}\frac{\mathrm{d}}{\mathrm{d}u}V(1/u)} \tag{3.22}$$

Solve for given $u = u_0$ for $\phi = \phi_0$.

$$\frac{\mathrm{d}u}{\mathrm{d}\phi} = 0 \quad \Longleftrightarrow \quad \text{max or min } r \text{ point.} \tag{3.23}$$

The absidal points are shown in Fig. 3.2.

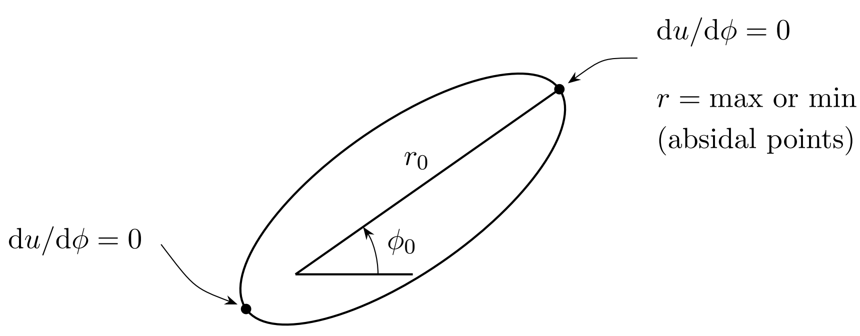

Figure 3.2: The absidal points.

3.7 Gravitation case. Kepler problem

$$V = -\frac{k}{r} = -ku, \qquad \frac{\mathrm{d}V}{\mathrm{d}u} = -k, \tag{3.24}$$

so Eq. (3.22) becomes

$$\boxed{\frac{\mathrm{d}^2 u}{\mathrm{d}\phi^2} + u = \frac{k\mu}{\ell^2}} \qquad \text{harmonic oscillator with shifted origin} \tag{3.25}$$

Aside. Put $\tilde{u} = u - k\mu/\ell^2$. Then

$$\boxed{\frac{\mathrm{d}^2 \tilde{u}}{\mathrm{d}\phi^2} + \tilde{u} = 0} \tag{3.26}$$

Homogeneous solution:

$$\tilde{u}(\phi) = A\cos(\phi - \phi_0), \tag{3.27}$$

$$\tilde{u}'(\phi) = -A\sin(\phi - \phi_0), \tag{3.28}$$

$$\tilde{u}''(\phi) = -A\cos(\phi - \phi_0) = -\tilde{u}(\phi). \tag{3.29}$$

Hence

$$\boxed{u(\phi) = \frac{k\mu}{\ell^2} + A\cos(\phi - \phi_0)}, \qquad u_0 = \frac{1}{r_0}, \tag{3.30}$$

which may be written

$$\boxed{u(\phi) = \frac{k\mu}{\ell^2}\Big[e\cos(\phi - \phi_0) + 1\Big] = \frac{1}{r}} \tag{3.31}$$

with e constant,

$$e \equiv \frac{A}{k\mu/\ell^2}. \tag{3.32}$$

3.8 Using the universal method

$$u'' = -u + \frac{k\mu}{\ell^2} \tag{3.33}$$

$$\Big\Downarrow \times\ u'(\phi)$$

$$\frac{\mathrm{d}u}{\mathrm{d}\phi} = \left(u'\right)^2 = \sqrt{\cdot \qquad\qquad - \qquad\qquad}$$

$$\mathrm{d}\phi = \frac{\mathrm{d}u}{u'}$$

$$\vdots$$

$$\boxed{u = \frac{\mu k}{\ell^2}\left[1 + \sqrt{1 + \frac{2\ell^2 E}{\mu k^2}}\ \cos(\phi - \phi_0)\right] = \frac{1}{r}} \tag{3.34}$$

$$\boxed{e \equiv \sqrt{1 + \frac{2\ell^2 E}{\mu k^2}}} \tag{3.35}$$